LESSON 3.2
EQUIVALENT CONVERSIONS: FRACTIONS PERCENTAGES – PART 1

LESSON INTRODUCTION

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.

LEARNING TARGETS

- I can convert between fractions with denominators of 100 and percentages.
- I can convert between fractions and percentages using equivalent fractions.

BENCHMARK COHERENCE

BUILDING ON

MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.

WORKING TOWARD

MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.

ESSENTIAL QUESTIONS

- What is a percentage?
- How do percentages relate to fractions?
- How do you convert fractions to percentages and percentages to fractions?

LESSON NARRATIVE

In this lesson, students learn how to convert between fractions with denominators of 100 and percentages, and how to convert between fractions and percentages using equivalent fractions. Students begin with a video on the history of the numerical system, then explore the meaning of percentage and how it relates to fractions with a denominator of 100. Students will then use this newly acquired knowledge to form equivalent fractions to percentages using multiplication and equivalent fractions.

COMPONENT SUMMARY

3.2.1 WARM-UP *(~5 MIN.)*

Place Value System Video

3.2.2 GUIDED INSTRUCTION *(10-15 MIN.)*

Exploration on percentages and relationship to fractions with a denominator of 100

3.2.3 GUIDED PRACTICE *(10-15 MIN.)*

Practice converting fractions to percentages using equivalent fractions

3.2.4 YOUR TURN! *(10-15 MIN.)*

Students practice forming equivalent values and converting fractions to percentages using equivalent fractions

3.2.5 WRAP-UP *(~5 MINUTES)*

Create equivalent fractions

RESOURCES

GLOSSARY

Key Terms:

- Percent

Additional Terms:

- Equivalent fractions
- Factors

MATERIALS

- Numerical Systems Video
- (Optional) Chart paper
- (Optional) Markers
- (Optional) 100-grids

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may believe a reduced fraction to be ‘new’ or ‘different’ than the original (or equivalent).
- Students may only perform the multiplication operation on the denominator to get it equal to 100.
- Students may interpret single digit percentages as tenths place as decimals (i.e. 2% = 0.2).

THINGS TO CONSIDER

- Highlight the equivalence of representations of fractions or decimals ($\frac{1}{2}$ is the same value as $\frac{5}{10}$).
- Consider offering factor trees and factor cards for students to find factors quickly, spending more time with the concept of equivalence instead of just on factoring.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

3.2.2 Guided Instruction and 3.2.3 Guided Practice position students to identify, compare, and contrast mathematical representations of fractions and percentages, and methods for conversion. Purposefully using words like “compare” or “connect” in discussion prompts students to make connections between prior knowledge and new concepts.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

3.2.2 Guided Instruction includes multiple entry points in representing fractions and percentages in different forms. Consider highlighting these representations and encouraging students to make connections or build their own models with manipulatives or other visual methods, showing how percentages and fractions can represent an equivalent value.

DIFFERENTIATE INSTRUCTION

Student choice, small-group learning, and leveling problems are all strategies to differentiate content for all students to engage. Consider one of these strategies for implementing Your Turn!

For visual learners, consider providing students with graph paper, 100-grids, or math manipulatives to visually represent converting between equivalent fractions and percentages in 3.2.2 Guided Instruction and 3.2.3 Guided Practice. This may also activate prior knowledge of fractions from previous grades.

Have students create an anchor chart using the problems from 3.2.4 Your Turn! on how to convert percentages and fractions instead of completing all the problems in this section. This activity may engage visual learners by helping them represent their understanding on paper while also engaging students with a choice in how they design their poster.

3.2.1 WARM-UP: A BRIEF HISTORY OF NUMERICAL SYSTEMS

Component Overview: *Play the "A Brief History of Numerical Systems" video about how the place value system was perfected by Indian mathematicians in the 8th century. After students watch this 5-minute video, prompt the students to read and answer the questions in the Warm-Up independently.*

QUESTIONING

Consider an extension project featuring the contribution of various cultures to the number system.

- India and the Base-10 System
- The Mayans and Math
- Mathematicians of the African Diaspora

3.2.1 Warm-Up: A Brief History of Numerical Systems

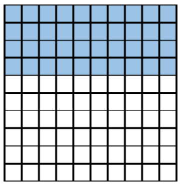
Humans went from using body parts, tally marks, and symbols to count objects, to what we now know as the base-ten system. Multiple civilizations around the world developed positional numbering systems, and both Mayan and Indian civilizations developed the concept of "zero." The base-ten system was perfected by eighth century Indian mathematicians. They realized that they could combine the concept of "zero" with a positional system that used repeatable symbols, whose values are determined by their position in a sequence. A base of ten was selected because, quite simply, we have ten fingers. Centuries later, this Hindu-Arabic numeral system began to spread to Europe and replaced the Roman numeral system. This is what we use today.

1. **What is the historical importance of the number zero?**
   Answers will vary. Zero helped form the base-ten system. Zero brought many cultures ideas about the number system together.

2. **If zero was never created, how would that affect how we do math today?**
   Answers will vary. Without zero, the base-ten system would not exist. Without zero, we would still use Roman Numerals.

3.2.2 Guided Instruction: Converting Between Fractions and Percentages

1. The model represents a numerical value. Use the model to answer the questions that follow.



   a. How many total squares are in the image?
   100

   b. How many of the squares are shaded blue?
   40

   c. What numerical value does the model represent, expressed as a fraction of $\frac{\text{blue shaded squares}}{\text{total squares}}$?
   $\frac{40}{100} = \frac{4}{10} = \frac{2}{5}$

Recall that a ***percentage*** is a ratio expressed as a fraction of 100.

> ***Percent*** (%) means "per 100," "for every 100," or "out of 100."

For example, 40% means "40 out of 100," so 40% is equivalent to the fraction $\frac{40}{100}$.

2. What decimal value has the same value as 40% and $\frac{40}{100}$?

Equivalent Values		
Percent Form	**Fraction Form**	**Decimal Form**
40%	$\frac{40}{100}$	0.40

3. With your partner, complete the table by rewriting each value in the equivalent form indicated.

Fraction Form	Percent Form
$\frac{50}{100}$	50%
$\frac{71}{100}$	71%
$\frac{2}{100}$	2%
$\frac{150}{100}$	150%
$\frac{42}{100}$	42%

When converting fractions to percentages, generate an equivalent fraction with a denominator of 100 when possible.

4. Complete the table.

Original Fraction	Equivalent Fraction with a Denominator of 100	Percentage
$\frac{6}{10}$	$\frac{6}{10} \times \frac{10}{10} = \frac{60}{100}$	60%
$\frac{23}{50}$	$\frac{23}{50} \times \frac{2}{2} = \frac{46}{100}$	46%
$\frac{1}{5}$	$\frac{1}{5} \times \frac{20}{20} = \frac{20}{100}$	20%

3.2.2 GUIDED INSTRUCTION: CONVERTING BETWEEN FRACTIONS AND PERCENTAGES

Component Overview: *This section guides students to understand the meaning of percentage and how it relates to fraction with a denominator of 100.*

DIFFERENTIATE INSTRUCTION

For visual learners, consider having a 100-grid paper printed and laminated (or multiple 100-grids on one paper) for students to practice shading in different percentages throughout this lesson. This entry point may help students make the connection by coloring in the value out of 100 that is being represented by the percentage.

TEACHER MOVES

Utilize Lesson 1 resources (definitions and video) for additional support in "percent" and "percentage" out of 100.

TEACHER MOVES

Compare and Connect

Position students to make connections between the "out of 100" forms in question 3 and original fractions in question 4.

- How is $\frac{6}{10}$ the same? How is it different?
- Why would $\frac{6}{10}$ not be 6%?
- What does that make you think about when converting fractions to percentages?

3.2.3 GUIDED PRACTICE: CONVERTING BETWEEN FRACTIONS AND PERCENTAGES

Component Overview: *In small groups, students convert between fractions with denominators and percentages, and convert between fractions and percentages using equivalent fractions.*

ENRICHMENT

Notice the first three rows compared to rows four and five. If students are stuck or unsure how to convert the percentage to a fraction, prompt with questioning:

- What do we know about percentages? (out of 100)
- What would be "out of 100" here?
- What does that look like?
- Can this be reduced? How?

3.2.3 Guided Practice: Converting Between Fractions and Percentages

1. Complete the table by rewriting each value in the equivalent form indicated. Simplify where applicable.

Fraction Form	Percent Form
$\frac{20}{50}$	40%
$\frac{13}{25}$	52%
$\frac{9}{10}$	90%
$\frac{30}{100} = \frac{3}{10}$	30%
$\frac{18}{100} = \frac{9}{50}$	18%
$\frac{5}{4}$	125%
$\frac{200}{100} = \frac{2}{1} = 2$	200%

3.2.4 Your Turn!

1. Rewrite each fraction as an equivalent percentage.

a. $\frac{17}{100}$ 17% b. $\frac{42}{50}$ 84% c. $\frac{36}{25}$ 144% d. $\frac{96}{100}$ 96%

e. $1\frac{9}{100}$ 109% f. $\frac{24}{20}$ 120% g. $\frac{1}{50}$ 2% h. $\frac{4}{5}$ 80%

2. Convert each to a fraction. Simplify where applicable.

a. 90% $\frac{9}{10}$ b. 167% $1\frac{67}{100}$ c. 86% $\frac{43}{50}$ d. 33% $\frac{33}{100}$

e. 226% $2\frac{13}{50}$ f. 29% $\frac{29}{100}$ g. 50% $\frac{1}{2}$ h. 18% $\frac{9}{50}$

3.2.4 YOUR TURN!

Component Overview: *Students convert between fractions and percentages and convert between percentages and fractions using equivalent fractions. Consider choosing specific questions based on learners' needs or having students use these problems to create their own anchor chart to summarize the processes and methods for converting fractions and decimals.*

DIFFERENTIATE INSTRUCTION

Consider strategically differentiating during Your Turn! to increase entry points for students.

- Choice: Increase student agency by allowing students to choose four fractions from question 1 and 4 percentages from question 2. Then they must find a classmate who completed each question and exchange check marks for correct answers until all have been verified.
- Small Group: Pull a small group based on observational data from the lesson for intervention while other students are practicing independently.
- Levels: Identify problems with low or high entry points and differentiate between "levels" in which students must "graduate" from level to level (i.e. $\frac{17}{100}$ would be Level 1, $\frac{42}{50}$ would be Level 2, and $\frac{24}{20}$ would be Level 3).,

3.2.5 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *Students use background knowledge on equivalent fractions and newly acquired knowledge of percentages to write three additional equivalent fractions to a given percentage. This open-ended Wrap-Up assesses students' ability to convert between fractions and percentages using equivalent fractions.*

QUESTIONING

For an added challenge, consider having students write as many equivalent fractions as they can think of (given a time limit). Consider adding a recognition or shout-out during the Warm-Up for the next lesson to the student who generated the most equivalent expressions.

3.2.5 Wrap-Up: Ticket Out the Door

1. Use the numbers below to create three more fractions equivalent to any three percentages in the table.

15	4	21	30	25	40
32	1	8	2	6	10
7	50	72	5	80	3

10%	20%	30%	40%	50%	60%	70%	80%	90%
$\frac{1}{10}$	$\frac{1}{5}$	$\frac{3}{10}$	$\frac{2}{5}$	$\frac{1}{2}$	$\frac{3}{5}$	$\frac{7}{10}$	$\frac{4}{5}$	$\frac{72}{80}$
$\frac{3}{30}$	$\frac{2}{10}$	$\frac{15}{50}$	$\frac{4}{10}$	$\frac{2}{4}$	$\frac{6}{10}$	$\frac{21}{30}$	$\frac{8}{10}$	
$\frac{4}{40}$	$\frac{3}{15}$		$\frac{6}{15}$	$\frac{3}{6}$	$\frac{15}{25}$		$\frac{32}{40}$	
$\frac{5}{50}$	$\frac{5}{25}$		$\frac{10}{25}$	$\frac{4}{8}$	$\frac{30}{50}$		$\frac{40}{50}$	
	$\frac{6}{30}$			$\frac{5}{10}$				
	$\frac{8}{40}$			$\frac{15}{30}$				
	$\frac{10}{50}$			$\frac{25}{50}$				

Answers will vary.